INTRODUCTION

This project presents a Fake News Detection System in which it uses Machine Learning and National Language Processing techniques to classify news as Real or Fake. In this project, it also includes a Graphical User Interface. The main of this project is to save people from misinformation and false news because detecting fake news manually is difficult and time consuming. As in today's world information runs rapidly through social media and other platforms.

OBJECTIVE

The main three objects of the project is -:

1-To develop a News Detection system like it can classify whether the given news is Real or Fake News.

2-To show and demonstrate the use of Machine Learning in the problems of real world.

3-To provide a simple and accessible interface for all the users and make a user friendly model.

PROBLEM STATEMENT

In this digital era of social media and internet, fake news can mislead the youth and future generation of the country which will create confusion and panic among the people due to which it sometimes can even cause harm. Because of this, there is a need for an automated system which help users to quickly verify the authenticity of the news and verify whether the news is Real or Fake.

SCOPE OF THE PROJECT

1 – Focuses on the text based news detection which is fake.

2 – Uses TF-IDF vectorization for the processing of text.

3 – Provides a Graphical User Interface using Tkinter.

4 – It implements logistic regression for the classification.

Limitations

1 – Uses a small dataset.

2 – It cannot handle images or videos.

3 – The accuracy may be limited due to very limited real world data.

TECHNOLOGIES USED

The technologies used in the project are-:

1 – The programming language used in the project is Python.

2 – The libraries used in the project is Tkinter for GUI and Scikit-learn for Machine Learning.

3 – The concepts which are used in the project are National Language Processing, TF-IDF vectorization and Logistic Regression.

SYSTEM ARCHITECTURE

Steps-:

1 – Input the text of the news from the user.

2 – Convert the text input by the user into numerical form using TF-IDF.

3 – Apply the trained Logistic Regression Model.

4 – Predict the output. The output can be “News is Real (Real News)” or “News Is Fake (Fake News)”.

5 – Display the result in the Graphical User Interface.

IMPLEMENTATION

The system is implemented in python by

1 – Backend- Machine Learning Model

2 – Frontend- Tkinter Graphical User Interface.

KEY COMPONENTS

1 – Input the field for the text of the news.

2 – Button for prediction.

3 – Label to Display the result.

USER INTERFACE

The Graphical User Interface includes - :

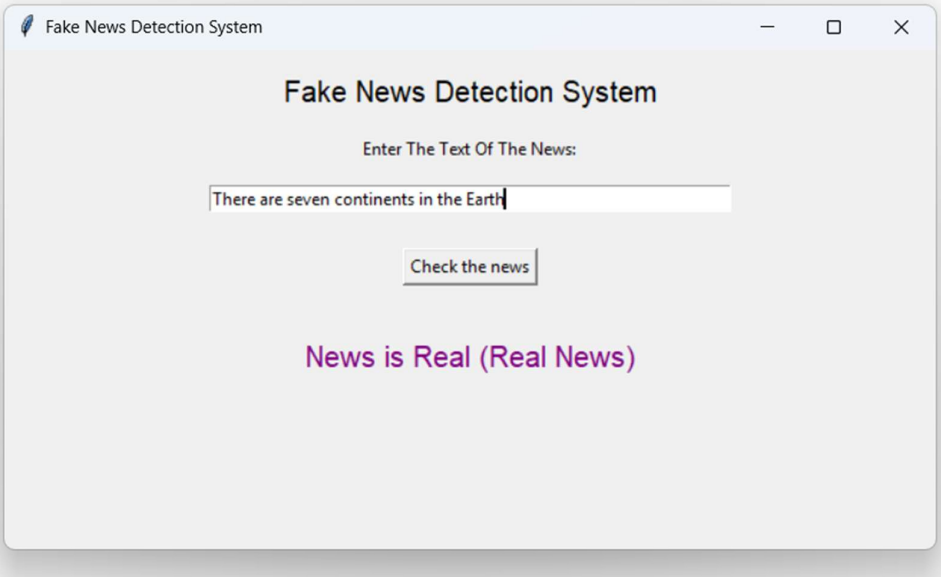
1 – The title which is Fake News Detection System.

2 – Input field for entering the text of the news.

3 – Button which is “Check the news”.

4- Output label which is used for showing the result.

Screenshots



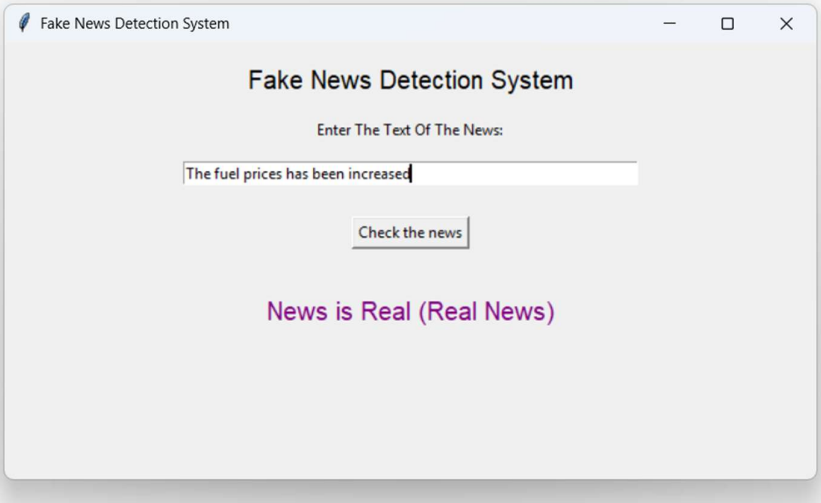
Fake News Detection System
Fake News Detection System
Enter The Text Of The News:
The account has been credited with 10000 rupees
Check the news
News is Fake (Fake News)

Fake News Detection System
Fake News Detection System
Enter The Text Of The News:
Click to get free mobile phone
Check the news
News is Fake (Fake News)

Fake News Detection System
Fake News Detection System
Enter The Text Of The News:
Technology improves everyone and helps everybody
Check the news
News is Real (Real News)

Fake News Detection System
Fake News Detection System
Enter The Text Of The News:
have to enter your pin after scanning the QR code to recieve money
Check the news
News is Fake (Fake News)

Fake News Detection System
Fake News Detection System
Enter The Text Of The News:
The cost of the newspaper is doubled from now on
Check the news
News is Fake (Fake News)


Fake News Detection System
Fake News Detection System
Enter The Text Of The News:
The fuel prices has been increased
Check the news
News is Real (Real News)

Fake News Detection System
Fake News Detection System
Enter The Text Of The News:
Check the news
Enter the text of the news

FUTURE ENHANCEMENTS

1 – It can use large real world dataset.

2 – It improves accuracy with advanced models.

3 – Add web interface or mobile interface.

4 – It includes news source verification.

APPLICATIONS

The applications are-:

1 – Tools for news verification.

2 – Education AI Projects.

3 – Social media content filtering.

CONCLUSION

This project demonstrates how Machine Learning and NLP can be used to address real- world problems like the detection of news which is fake. Although simple, it provides a strong foundation for building more advanced and accurate systems in the future.

REFERENCES

1 – VITYARTHI MODULES

2 – Scikit-learn Documentation

3 – Python Official Documentation

4 – Tutorials for Machine Learning

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